



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Symbiosis International (Deemed University)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 09

Subject:	DevOps Lab
TCODE:	TE7950
Name of Student	Ria Vinod
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Branch	CSE, Batch (2023-27), L3
Academic Year & Semester	25 – 26, Sem VII
Date of Performance	23 rd July, 2026
Title of Assignment:	Container Management & Networking
Theory:	Docker container management is the process of creating, running, monitoring, stopping, inspecting, and removing containers. Port mapping allows applications running inside a container to be accessed from the host machine by mapping a host port to a container port. Docker provides commands such as docker ps to view running containers, docker inspect to examine detailed container configuration and network settings, docker stop to gracefully stop a container, and docker rm to remove unused containers. These commands help efficiently manage the lifecycle of Docker containers and verify their network configuration and status. Commands Used • Created and started the Docker container in detached mode with port mapping: docker run -d -p 8081:5000 --name flask-network-demo my-flask-app • Listed all running Docker containers: docker ps • Stopped the running Docker container: docker stop flask-network-demo • Listed all Docker containers, including stopped containers: docker ps -a • Inspected the container's network configuration and settings: docker inspect flask-network-demo • Removed the stopped Docker container: docker rm flask-network-demo

- Verified that the container had been removed:
docker ps -a

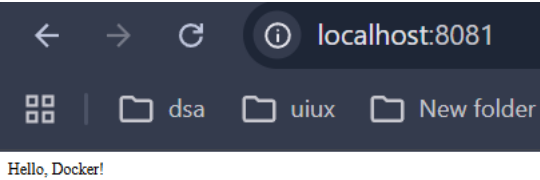
Implementation:

Step 1: Run a Container with Port Mapping

```
PS C:\Users\riavi\my-flask-app> docker run -d -p 8081:5000 --name flask-network-demo my-flask-app
c738bd0510ee356a047bba588d1497d30b2fddd39f7b6acfb0afd3e27e1c396b
PS C:\Users\riavi\my-flask-app>
```

Started Docker Container — Started the Docker container named flask-network-demo in detached mode and mapped host port 8081 to container port 5000, allowing the application to run in the background.

Step 2: Access the Application via Your Browser



Verified Application Access — Accessed the Dockerized Flask application through http://localhost:8081 and verified that the application was successfully running inside the Docker container.

Step 3: List Running Containers

```
PS C:\Users\riavi\my-flask-app> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED
STATUS        PORTS          NAMES
c738bd0510ee   my-flask-app   "python app.py"         2 minutes ago
Up 2 minutes   0.0.0.0:8081->5000/tcp, [::]:8081->5000/tcp   flask-network-demo
defd1b9f4c1a   jenkins/jenkins:lts-jdk17 "/usr/bin/tini -- /u..." 6 days ago
Up 5 hours     0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp   myjenkins
PS C:\Users\riavi\my-flask-app>
```

Listed Running Containers — Displayed all currently running Docker containers using the docker ps command and verified that the flask-network-demo container was running with port mapping from 8081 on the host to 5000 inside the container.

Step 4: Stop the Running Container

```
PS C:\Users\riavi\my-flask-app> docker stop flask-network-demo
PS C:\Users\riavi\my-flask-app>
```

Stopped Docker Container — Stopped the running flask-network-demo container using the docker stop command, terminating the application gracefully.

Step 5: List Running and All Containers

```

PS C:\Users\riavi\my-flask-app> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
defd1b9f4c1a   jenkins/jenkins:lts-jdk17          "/usr/bin/tini -- /u..." 6 days ago    Up 5 hours    0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp
myjenkins

PS C:\Users\riavi\my-flask-app> docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
c738bd0510ee   my-flask-app                        "python app.py"          7 minutes ago  Exited (0)
2 minutes ago
flask-network-demo
ac7bba30e105   my-flask-app                        "python app.py"          34 minutes ago Exited (0)
10 minutes ago
festive_morse
5195a83ad0e8   785e6affefd7                        "python app.py"          37 minutes ago Exited (1)
37 minutes ago
relaxed_williams
defd1b9f4c1a   jenkins/jenkins:lts-jdk17          "/usr/bin/tini -- /u..." 6 days ago    Up 5 hours    0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp
myjenkins
63d8b2cf90b8   hello-world                          "/hello"                  6 days ago    Exited (0)
6 days ago
funny_dirac

PS C:\Users\riavi\my-flask-app>

```

Verified Container Status — Used the `docker ps -a` command to display all Docker containers and verified that the `flask-network-demo` container had stopped successfully and was in the Exited state.

Step 6: Inspect the Container's Network Settings

```

PS C:\Users\riavi\my-flask-app> docker inspect flask-network-demo
[
  {
    "NetworkSettings": {
      "SandboxID": "",
      "SandboxKey": "",
      "Ports": {},
      "Networks": {
        "bridge": {
          "IPAMConfig": null,
          "Links": null,
          "Aliases": null,
          "DriverOpts": null,
          "GwPriority": 0,
          "NetworkID": "ae7b1c98448c112eee0784f77a5090772b740ff5be9fedf421799e0803a45700",
          "EndpointID": "",
          "Gateway": "",
          "IPAddress": "",
          "MacAddress": "",
          "IPPrefixLen": 0,
          "IPv6Gateway": "",
          "GlobalIPv6Address": "",
          "GlobalIPv6PrefixLen": 0,
          "DNSNames": null
        }
      }
    }
  }
]

```

Inspected Container Network Settings — Used the `docker inspect` command to view detailed information about the `flask-network-demo` container, including its `NetworkSettings` and network configuration.

Step 7: Remove the Stopped Container

```

PS C:\Users\riavi\my-flask-app> docker rm flask-network-demo
flask-network-demo
PS C:\Users\riavi\my-flask-app> docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS
PORTS         NAMES
ac7bba30e105   my-flask-app   "python app.py"         38 minutes ago Exited (0)
14 minutes ago festive_morse
5195a83ad0e8   785e6affefd7   "python app.py"         41 minutes ago Exited (1)
41 minutes ago relaxed_williams
defd1b9f4c1a   jenkins/jenkins:lts-jdk17 "/usr/bin/tini -- /u... 6 days ago    Up 5 hours
0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:5 myjenkins
0000->50000/tcp
63d8b2cf90b8   hello-world    "/hello"                 6 days ago    Exited (0)
6 days ago
funny_dirac
PS C:\Users\riavi\my-flask-app>

```

Removed Docker Container — Removed the flask-network-demo container using the docker rm command and verified its successful removal by confirming that it no longer appeared in the list of Docker containers.

Subject Faculty Name: Prof. Arijit Dutta